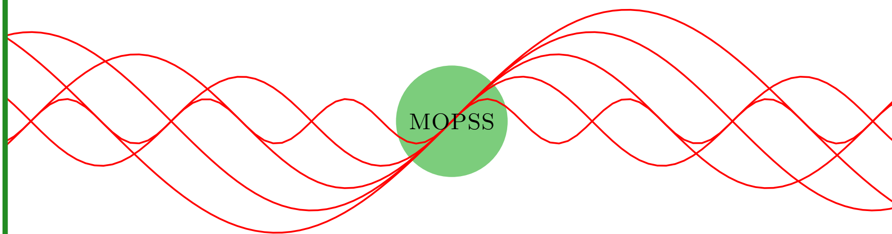
Finite differences

MOPSS

Mathematics Olympiad

Problem Solving Sessions



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https://jpsaha.github.io/MOTP/MOPSS/

Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

§1 Finite differences

Note that

$$k^2 - (k-1)^2 = 2k - 1$$

holds for any integer k. In particular, given a positive integer n, we have

$$\begin{aligned}
2^2 - 1^2 &= 2 \cdot 2 - 1, \\
3^2 - 2^2 &= 2 \cdot 3 - 1, \\
4^2 - 3^2 &= 2 \cdot 4 - 1, \\
\cdots &= \cdots, \\
n^2 - (n-1)^2 &= 2 \cdot n - 1.
\end{aligned}$$

Adding them, it follows that

$$n^2 - 1 = 2(2 + 3 + \cdots + n) - (n-1),$$

which yields

$$\boxed{1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}}.$$

Consider the polynomial $P(x) = x^3$. Note that

$$P(k) - P(k-1) = 3k^2 - 3k + 1.$$

Using a similar argument as above, it follows that given a positive integer n, we have

$$\begin{aligned}
1^3 - 0^3 &= 3 \cdot 1^2 - 3 \cdot 1 + 1, \\
2^3 - 1^3 &= 3 \cdot 2^2 - 3 \cdot 2 + 1, \\
3^3 - 2^3 &= 3 \cdot 3^2 - 3 \cdot 3 + 1, \\
4^3 - 3^3 &= 3 \cdot 4^2 - 3 \cdot 4 + 1, \\
\cdots &= \cdots, \\
n^3 - (n-1)^3 &= 3 \cdot n^2 - 3 \cdot n + 1.
\end{aligned}$$

Adding them, it follows that

$$n^3 = 3(1 + 2^2 + 3^2 + \cdots + n^2) - 3(1 + 2 + 3 + \cdots + n) + n,$$

which yields

$$\begin{aligned}
1 + 2^2 + 3^2 + \cdots + n^2 &= \frac{n^3 - n}{3} + (1 + 2 + 3 + \cdots + n) \\
&= \frac{n^3 - n}{3} + \frac{n(n+1)}{2} \\
&= \frac{1}{6} n(n+1)(2n+1).
\end{aligned}$$

Example 1.1 (India RMO 2013b P3)**.** Consider the expression

$$2013^2 + 2014^2 + 2015^2 + \cdots + n^2.$$

Prove that there exists a natural number $n > 2013$ for which one can change a suitable number of plus signs to minus signs in the above expression to make the resulting expression equal 9999.

Summary — "Differentiating" a polynomial enough times makes it linear.

Walkthrough —

(a) Consider the polynomial $P(k) = k^2$, and the polynomial $Q(k) := P(k) - (k-1)$.

(b) Since $Q(k)$ is a linear polynomial in k, the difference $R(k) := Q(k) - Q(k-2)$ is a constant, that is, it does not depend on k.

(c) Note that $R(k)$ is a **± 1-linear combination**[a] of four consecutive squares.

(d) Does this help?

[a]What does it mean?

Solution 1. Consider the polynomial $P(k) = k^2$, and the polynomial $Q(k) := P(k) - (k-1)$. Since $Q(k)$ is a linear polynomial in k, the difference $R(k) := Q(k) - Q(k-2)$ is a constant, that is, it does not depend on k. Indeed, $Q(k) = 2k - 1$, and $R(k) = 4$. Note that $R(k) = k^2 - (k-1)^2 - (k-2)^2 + (k-3)^2$.

Note that

$$2013^2 + 2014^2 + 2015^2 + 2016^2 + 2017^2 > 9999$$

holds, and the integers $2013^2 + 2014^2 + 2015^2 + 2016^2 + 2017^2, 9999$ are congruent modulo 4, that is, they differ by a multiple of 4. Let $m \geq 1$ be an integer such that

$$9999 = 2013^2 + 2014^2 + 2015^2 + 2016^2 + 2017^2 - 4m$$

holds. Since

$$-k^2 + (k+1)^2 + (k+2)^2 - (k+3)^2 = -4,$$

it follows that

$$\begin{aligned}
&9999\\
&= 2013^2 + 2014^2 + 2015^2 + 2016^2 + 2017^2\\
&- 2018^2 + 2019^2 + 2020^2 - 2021^2\\
&- \dots\\
&- \big((2018 + 4(m-1))^2 + (2019 + 4(m-1))^2\\
&\qquad + (2020 + 4(m-1))^2 - (2021 + 4(m-1))^2\big).
\end{aligned}$$

It follows that there exists a natural number $n = 2021 + 4(m-1) > 2013$, for which one can change a suitable number of plus signs to minus signs in the expression

$$2013^2 + 2014^2 + 2015^2 + \cdots + n^2$$

to make the resulting expression equal to 9999. ∎

Exercise 1.2 (Bay Area MO 12 2016 P4). Find a positive integer N and $a_1, a_2, \dots, a_N$, where $a_k = 1$ or $a_k = -1$ for each $k = 1, 2, \dots, N$, such that

$$a_1 \cdot 1^3 + a_2 \cdot 2^3 + a_3 \cdot 3^3 + \cdots + a_N \cdot N^3 = 20162016,$$

or show that this is impossible.

Summary — "Differentiating" a polynomial enough times makes it linear.

Walkthrough —

(a) Consider the polynomial $P(k) := k^3$. Note that $R(k) := P(k) - P(k-1)$ is a quadratic polynomial in k.

(b) Also note that $S(k) := R(k) - R(k-2)$ is a linear polynomial in k.

Solution 2. Consider the polynomial $P(k) := k^3$. Note that $R(k) := P(k) - P(k-1)$ is equal to $3k^2 - 3k + 1$. Also note that $S(k) := R(k) - R(k-2)$ is equal to $6(2k-2) - 6$. This gives $S(k) - S(k-4) = 48$. It follows that **some ± 1-linear combination** of any given eight consecutive cubes is equal to 48. More specifically,

$$k^3 - (k-1)^3 - (k-2)^3 + (k-3)^3 - (k-4)^3 + (k-5)^5 + (k-6)^3 - (k-7)^3 = 48,$$

or equivalently,

$$-k^3 + (k+1)^3 + (k+2)^3 - (k+3)^3 - (k+4)^3 + (k+5)^3 + (k+6)^3 - (k+7)^3 = 48.$$

Note that 20162016 is divisible by 3 and 16. Since $3, 16$ do not have any common prime factor, it follows that 20162016 is a multiple of 48. Write

$$f(k) = -k^3+(k+1)^3+(k+2)^3-(k+3)^3-(k+4)^3+(k+5)^3+(k+6)^3-(k+7)^3.$$

Note that

$$f(1) + f(9) + f(17) + \cdots + f(8m-7) = 20162016,$$

where m denotes the integer $20162016/48$. We conclude that one may take $N = 8m = 20162016/6 = 3360336$ so that the given condition holds. ■

Consider the finite difference operator Δ defined on the set of polynomials by

$$\Delta P(x) = P(x+1) - P(x).$$

For a polynomial $P(x)$, we define the r-th finite difference recursively as

$$\Delta^r P(x) = \Delta(\Delta^{r-1} P(x)),$$

with the convention that $\Delta^0 P(x) = P(x)$.

For an integer $k \geq 0$, define the binomial coefficient $\binom{x}{k}$ as

$$\binom{x}{k} = \frac{x(x-1)(x-2)\cdots(x-k+1)}{k!}$$

if $k \geq 1$, and $\binom{x}{0} = 1$. Note that $\binom{x}{k}$ is a polynomial in x of degree k.

Theorem 1

Let $P(x)$ be a polynomial of degree n with leading coefficient a. Then the n-th finite difference $\Delta^n P(x)$ is a constant polynomial and is equal to $a \cdot n!$, and the polynomial $P(x)$ can be expressed as

$$P(x) = \sum_{k=0}^{n} \Delta^k P(0) \binom{x}{k}.$$

Exercise 1.3 (Harvard-MIT Mathematics Tournament 2015 P1)**.** The complex numbers x, y, z satisfy

$$\begin{aligned} xyz &= -4, \\ (x+1)(y+1)(z+1) &= 7, \\ (x+2)(y+2)(z+2) &= -3. \end{aligned}$$

Find, with proof, the value of $(x+3)(y+3)(z+3)$.

Walkthrough —

(a)

Solution 3. Let $P(t)$ denote the polynomial $(t+x)(t+y)(t+z)$. Note that

$$P(0) = -4, P(1) = 7, P(2) = -3.$$

This gives

$$\Delta P(0) = 11, \Delta P(1) = -10, \Delta^2 P(0) = -21, \Delta^3 P(0) = 6.$$

It follows that

$$\Delta^2 P(1) = \Delta^2 P(0) + \Delta^3 P(0) = -15,$$

which in turn gives

$$\Delta P(2) = \Delta P(1) + \Delta^2 P(1) = -25.$$

Finally, we have

$$P(3) = P(2) + \Delta P(2) = -28.$$

■

Example 1.4. Let n be an even positive integer, and P be a polynomial of degree n. Assume that $P(0) = 1$ and $P(j) = 2^{j-1}$ for all $j = 1, 2, \dots, n$. Prove that

$$P(n+2) = 2P(n+1) - 1.$$

Solution 4. Note that it suffices to prove the following claim.

Claim — Let n be an even positive integer, and P be a polynomial of degree n satisfying

$$P(0) = 1, P(j) = 2^{j-1}$$

for all $j = 1, 2, \dots, n$. Then the polynomial P satisfies

$$P(n+1) = 2^n, P(n+2) = 2^{n+1} - 1.$$

Proof. Let us first consider the case $n = 2$. Using

$$P(0) = 1, P(1) = 1, P(2) = 2,$$

we have

$$\Delta P(0) = 0, \Delta P(1) = 1,$$

and thus

$$\Delta^2 P(0) = 1.$$

Since P is a polynomial of degree 2, it follows that $\Delta^2 P(x)$ is a constant polynomial, and thus $\Delta^2 P(x) = 1$ for all integer x. Therefore, we have

$$\Delta P(2) = \Delta P(1) + \Delta^2 P(1) = 1 + 1 = 2,$$

and

$$\Delta P(3) = \Delta P(2) + \Delta^2 P(2) = 2 + 1 = 3.$$

This yields

$$P(3) = P(2) + \Delta P(2) = 2 + 2 = 4,$$

and

$$P(4) = P(3) + \Delta P(3) = 4 + 3 = 7.$$

This proves [1] the claim for $n = 2$.

Assume that the claim holds for some even positive integer k. Let us consider the case $n = k + 2$. Let P be a polynomial of degree $k + 2$ such that

$$P(0) = 1, P(j) = 2^{j-1}$$

for all $j = 1, 2, \ldots, k + 2$. Note that ΔP takes the values

$$0, 1, 2, 4, \ldots, 2^k$$

at $0, 1, 2, 3, \ldots, k + 1$, respectively. This implies that $\Delta^2 P$ takes the values

$$1, 1, 2, 4, \ldots, 2^{k-1}$$

at $0, 1, 2, 3, \ldots, k$, respectively. Note that $\Delta^2 P$ is a polynomial of degree k. By the induction hypothesis, we have

$$\Delta^2 P(k + 1) = 2^k, \Delta^2 P(k + 2) = 2^{k+1} - 1.$$

This implies that

$$\Delta P(k + 2) = \Delta P(k + 1) + \Delta^2 P(k + 1) = 2^k + 2^k = 2^{k+1},$$

and

$$\Delta P(k + 3) = \Delta P(k + 2) + \Delta^2 P(k + 2) = 2^{k+1} + (2^{k+1} - 1) = 2^{k+2} - 1.$$

Therefore, we have

$$P(k + 3) = P(k + 2) + \Delta P(k + 2) = 2^{k+1} + 2^{k+1} = 2^{k+2},$$

[1] It is easier to **visualise** this using the following difference table:

x	0	1	2	3	4
$P(x)$	1	1	2	$2+2$	$2+2+3$
$\Delta P(x)$	0	1	2	3	
$\Delta^2 P(x)$	1	1	1		

and

$$P(k+4) = P(k+3) + \Delta P(k+3) = (2^{k+2}) + (2^{k+2} - 1) = 2^{k+3} - 1.$$

This proves the claim for $n = k + 2$. By induction, the claim holds for all even positive integers n. □

■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)